

JONAS SPINNER

CURRICULUM VITAE

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RESEARCH INTERESTS

- ML for particle physics
 - Generative modelling
 - Geometric deep learning
- Solving technical problems at the LHC and beyond
Flow matching, autoregressive transformers, finetuning strategies
Lorentz-equivariant graph networks and transformers

EXPERIENCE

Feodor Lynen Fellow IPPP, Durham University 10/2025 – Exp. 09/2028

- Topics: Physics-informed ML, ML for the LHC

EDUCATION

PhD in Physics Heidelberg University magna cum laude 10/2022 – 09/2025

- Thesis: Symmetry-aware networks for the LHC (advisor: Tilman Plehn)
- Topics: ML for the LHC, generative modelling, Lorentz-equivariant networks

MSc in Physics Karlsruhe Institute for Technology 1.1* 10/2020 – 09/2022

- Thesis: New Light Particles in Astrophysics, Cosmology and at Colliders (advisor: Ulrich Nierste)
- Visited IPPP, Durham University for 6 months during the MSc Thesis

BSc in Physics Karlsruhe Institute for Technology 1.1* 10/2017 – 09/2020

- Thesis: Dimension 7 operators in Tritium Beta Decay (advisor: Ulrich Nierste)

Abitur Robert-Gerwig Gymnasium Hausach 1.0* 06/2017

*German grading scale: from 1.0 (best) to 6.0 (worst)

TEACHING

Co-supervision MSc thesis Sebastian Pitz 07/2024 – Exp. 06/2024

- Topic: Jet Tagging with Learned Lorentz Equivariant Frames

Lecturer at Erum Datahub Active Training Course on Advanced Deep Learning 05/2024

- Delivered a 90-minute lecture on Transformers for 50 students

Head teaching assistant for Master-level course Machine Learning and Physics 10/2023 – 03/2024

- Organization of exercises and exams for 120 students

Co-supervision BSc thesis Nathanael Ediger 06/2023-10/2023

- Topic: LHC Event Generation with JetGPT - From variable orderings to joint training

Teaching assistant for 7 one-term courses in Theoretical Physics 04/2019 – 09/2023

- Discussions and marking for groups of 10-20 students

WORKSHOPS AND CONFERENCES

NeurIPS 2025	San Diego, US	12/2025
ACAT 2025	Hamburg, Germany	09/2025
MIAPbP Workshop	Munich, Germany	09/2025
IAIFI Summer Workshop	Boston, US	08/2025
NeurIPS 2024	Vancouver, Canada	12/2024
ML4Jets 2024	Paris, France	11/2024
PHYSTAT – Statistics meets Matching Learning	London, United Kingdom	09/2024
EuCAIFCON 2024	Amsterdam, Netherlands	04/2024
ML4Jets 2023	Hamburg, Germany	11/2023

DOCTORATE SCHOOLS

TASI 2024 - The Frontiers of Particle Theory	Boulder, Colorado, US	06/2024
Machine Learning in Particle Theory - MITP Summer School	Mainz, Germany	06/2023
Active Training Course on Advanced Deep Learning	Meinerzhagen, Germany	11/2022

SKILLS

Technical Languages Python, PyTorch, git, Wolfram Mathematica
German (native), English (fluent)

AWARDS

- Feodor Lynen Fellowship
- Prestigious German Studienstiftung scholarship (top 0.5% of all German students)
- Deutschlandstipendium scholarship (awarded to top-performing students across Germany)

PUBLICATIONS

- [1] A. Petitjean, T. Plehn, J. Spinner and U. Köthe, *Economical Jet Taggers – Equivariant, Slim, and Quantized* (2025), [arXiv:2512.17011](#)
- [2] A. Petitjean, A. Butter, K. Greif, S. Palacios Schweitzer, T. Plehn, J. Spinner and D. Whiteson, *Generative Unfolding of Jets and Their Substructure* (2025), [arXiv:2510.19906](#)
- [3] H. Bahl, S. Diefenbacher, N. Elmer, T. Plehn and J. Spinner, *Forecasting Generative Amplification* (2025), [arXiv:2509.08048](#)
- [4] L. Favaro, G. Gerhartz, F. A. Hamprecht, P. Lippmann, S. Pitz, T. Plehn, H. Qu and J. Spinner, *Lorentz-Equivariance without Limitations* (2025), [arXiv:2508.14898](#)
- [5] C. A. Manzari, J. Martin Camalich, J. Spinner and R. Ziegler, *The SN 1987A Cooling Bound on Dark Matter Absorption in Electron Targets* (2025), [arXiv:2508.00725](#)
- [6] J. S. Spinner, *Symmetry-aware networks for the LHC*, Ph.D. thesis, U. Heidelberg, [doi:10.11588/heidok.00036846](#) (2025)
- [7] J. Spinner, L. Favaro, P. Lippmann, S. Pitz, G. Gerhartz, T. Plehn and F. A. Hamprecht, *Lorentz Local Canonicalization: How to Make Any Network Lorentz-Equivariant*, In *39th conference on Neural Information Processing Systems* (2025), [arXiv:2505.20280](#)
- [8] A. Butter, F. Charton, J. M. Villadamigo, A. Ore, T. Plehn and J. Spinner, *Extrapolating Jet Radiation with Autoregressive Transformers*, *SciPost Phys.* **20**, 004 (2026), [doi:10.21468/SciPostPhys.20.1.004](#), [arXiv:2412.12074](#)
- [9] J. Brehmer, V. Bresó, P. de Haan, T. Plehn, H. Qu, J. Spinner and J. Thaler, *A Lorentz-equivariant transformer for all of the LHC*, *SciPost Phys.* **19**, 108 (2025), [doi:10.21468/SciPostPhys.19.4.108](#), [arXiv:2411.00446](#)
- [10] J. Spinner, V. Bresó, P. de Haan, T. Plehn, J. Thaler and J. Brehmer, *Lorentz-Equivariant Geometric Algebra Transformers for High-Energy Physics*, In *38th conference on Neural Information Processing Systems* (2024), [arXiv:2405.14806](#)
- [11] C. A. Manzari, J. Martin Camalich, J. Spinner and R. Ziegler, *Supernova limits on muonic dark forces*, *Phys. Rev. D* **108**, 103020 (2023), [doi:10.1103/PhysRevD.108.103020](#), [arXiv:2307.03143](#)
- [12] A. Butter, N. Huetsch, S. Palacios Schweitzer, T. Plehn, P. Sorrenson and J. Spinner, *Jet diffusion versus JetGPT – Modern networks for the LHC*, *SciPost Phys. Core* **8**, 026 (2025), [doi:10.21468/SciPostPhysCore.8.1.026](#), [arXiv:2305.10475](#)
- [13] M. Bauer, G. Rostagni and J. Spinner, *Axion-Higgs portal*, *Phys. Rev. D* **107**, 015007 (2023), [doi:10.1103/PhysRevD.107.015007](#), [arXiv:2207.05762](#)